

# **ATtiny85-Sample**

8-bit AVR Microcontroller with 8K Bytes In-System Programmable Flash

## Features

- High Performance, Low Power AVR 8-Bit Microcontroller
- 8K Bytes of In-System Self-Programmable Flash
- Real Time Counter with Separate Oscillator

## 7. Absolute Maximum Ratings

Storage Temperature ..... -65C to +150C  
Voltage on any Pin except RESET ..... -0.5V to VCC+0.5V  
Maximum Operating Voltage ..... 6.0V  
DC Current per I/O Pin ..... 40.0 mA

Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device.

## 8. Recommended Operating Conditions

Supply Voltage (VCC) ..... 2.7V to 5.5V

Operating Temperature ..... -40C to +85C

Maximum Operating Frequency at 5.5V ..... 20 MHz

## 9. Power Supply Decoupling

A 100 nF ceramic decoupling capacitor should be placed as close as possible to each VCC pin to minimize supply noise picked up on the power lines. An additional 10 uF bulk electrolytic or tantalum capacitor should be placed near the voltage regulator output.

## 10. Reset Circuit

The RESET pin has an internal pull-up resistor enabled by default. If the reset function is disabled in the fuse bits, an external 10 kOhm pull-up resistor to VCC should be added to the RESET pin to prevent spurious resets from noise.

## 11. Clock and Oscillator

When driving the device from an external crystal, a load capacitance of 12 pF to 22 pF is typically required, depending on the crystal manufacturer's specification. See Table 11-1 for recommended capacitor values for common crystal frequencies.

## 12. Layout Considerations

Keep decoupling capacitors as close as possible to the supply pins they protect. Minimize trace length between the crystal and the XTAL1/XTAL2 oscillator pins to reduce parasitic capacitance and improve oscillator stability.

## 13. Typical Application Circuit

Figure 13-1 shows a typical application circuit for the device including the recommended decoupling network and reset circuit described in the preceding sections.